Laser Properties

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Abstract

This experiment focused on characterizing the fundamental properties of a Helium-Neon (He-Ne) laser, specifically its degree of polarization, beam profile, divergence, and longitudinal mode spacing. Measurements were performed using a rotating linear polarizer, a micrometer-driven knife-edge for beam profiling, and a Fabry-Pérot interferometer to resolve spectral clusters. We obtained a degree of polarization of $\rho = 0.9985 \pm 0.0012$ and an experiental beam divergence of of 1.15 mrad, which is consistent with the theoretically calculated value of approximately 1.21 mrad.]Furthermore, the Gaussian beam waist was measured to be $D_0 = 0.272$ mm with an observed longitudinal mode spacing consistent with a cavity length of 2 cm. These results confirm that the laser operates as a high-purity TEM_{00} source that adheres closely to theoretical Gaussian propagation and Malus' Law.

1 Introduction

The unique properties of laser light, specifically its high degree of spatial coherence, mono-chromaticity, and collimation, make it a very useful tool in precision optical experiements and fundamental physics. This experiment aims to characterize a commercial He-Ne laser to verify its performance against theoretical models of electromagnetic wave propagation and cavity resonance. Understanding the spatial profile and polarization purity of the beam is critical for applications ranging from interferometry to fiber-optic coupling, where beam quality directly impacts system efficiency.

The background physics for this characterization relies on several key models. First, the polarization state is examined via Malus' Law, which describes the transmitted intensity through a polarizer as a function of the angle θ:

$$I(\theta) = I_{max} \cos^2(\theta - \theta_0) + I_{min} \tag{1}$$

where I_{min} accounts for background noise and finite extinction ratios. Spatially, the laser is expected to emit a TEM_{00} mode, characterized by a Gaussian intensity profile:

$$I(r) = I_0 \exp\left(\frac{-2r^2}{w^2}\right) \tag{2}$$

where w is the beam radius at which the field amplitude falls to $1/e$ of its axial value]. In this lab, we employ the knife-edge method to measure this profile. By recording the integrated power $P(x)$ as a razor blade translates across the beam, we fit the data to an error function:

$$P(x) = \frac{A}{2}\left[1 + \text{erf}\left(\frac{\sqrt{2}(x - x_0)}{w}\right)\right] \tag{3}$$

This allow us to determine the beam width w at various distances z to calculate divergence and the beam waist D_0. Finally, the spectral properties are governed by the cavity length L, which determines the longitudinal mode spacing or Free Spectral Range (FSR):

$$\Delta\nu = \frac{c}{2L} \tag{4}$$

By measuring these parameters, we can provide a complete profile of the laser's performance and verify the diffraction-limited nature of its propagation.

2 Experimental Methods

The experimental characterization of the He-Ne laser was divided into three primary segments: polarization analysis, spatial beam profiling, and spectral mode observation. The apparatus was mounted on a stabilized optical rail to ensure precise axial alignment between the laser aperture, various optical components, and the thermal power detector.

2.1 Polarization and Malus' Law

To determine the degree of polarization, a linear polarizer was placed in the beam path at the 30 cm mark on the rail. A thermal power meter, located at the 60 cm mark, recorded the transmitted intensity as the polarizer was rotated through 360° in 10° increments. To account for electronic drift and ambient noise, we performed a background calibration by recording the ambient power with the laser deactivated (0.95 μW), which was subsequently subtracted from all raw measurements. Each data point was averaged over a 10-second interval to minimize the impact of transient fluctuations in the laser power meter readings.

2.2 Gaussian Beam Profiling

Spatial beam characterization was performed using the knife-edge technique. A razor blade mounted on a micrometer-driven translation stage was used to progressively block the beam. As the blade moved through the beam, the integrated power was recorded by the sensor until the beam was fully blocked. This procedure was repeated at multiple axial distances (z) from the laser aperture to observe beam propagation. To improve signal-to-noise ratios and eliminate statistical errors from diffuse laser light, an iris was introduced in later trials to isolate the central TEM_{00} mode.

2.3 Interferometry and Longitudinal Modes

The spectral composition of the laser was investigated using a Fabry-Pérot (FB) interferometer. The signal path consisted of the laser beam entering the interferometer, with the resulting interference pattern captured by a sensor and displayed on a Tektronix oscilloscope. We adjusted the horizontal scale to 5 ms/div and calibrated the trigger level to resolve distinct clusters of longitudinal mode peaks. Traces were recorded at various polarization angles (0°, 90°, 180°, and 270°) to observe the relationship between polarization and mode structure.

3 Raw Data

3.1 Polarization Measurements

The transmitted power was recorded through a linear polarizer as a function of the rotation angle θ.]A background ambient power of 0.95 μW was recorded with the laser off and subtracted during the analysis phase.

Table 1: Transmitted Power vs. Polarizer Angle (Background Subtracted)

Angle ($\theta \pm 0.5°$)	**Power** ($P_{avg} \pm \sigma_P$ μ**W**)	**Angle** ($\theta \pm 0.5°$)	**Power** ($P_{avg} \pm \sigma_P$ μ**W**)
0°	71.4 ± 0.2	100°	311.0 ± 0.1
10°	31.0 ± 0.1	110°	333.0 ± 0.1
20°	5.6 ± 0.1	120°	340.0 ± 0.1
30°	2.4 ± 0.2	130°	328.5 ± 0.5
40°	18.3 ± 0.1	140°	293.0 ± 0.1
50°	55.6 ± 0.1	150°	245.0 ± 0.1
60°	106.5 ± 0.5	160°	185.0 ± 0.1
70°	158.0 ± 0.1	170°	122.0 ± 0.1
80°	208.5 ± 0.5	180°	69.4 ± 0.2
90°	268.0 ± 0.1		

3.2 Knife-Edge Spatial Beam Profiling

The beam profile was initially mapped at an axial distance of $z = 0.05$ m from the laser aperture using a micrometer-driven razor blade.

3.3 Gaussian Beam Propagation

The beam diameter was measured at various distances z from the lens or aperture to characterize the divergence and the beam waist position. These measurements utilize the 97.72% and 2.28%

Table 2: Knife-Edge Power Distribution at $z = 0.05$ m

Pos. ($x \pm 0.01$ **mm**)	**Power** ($P_{avg} \pm \sigma_P$ μ**W**)	**Pos.** ($x \pm 0.01$ **mm**)	**Power** ($P_{avg} \pm \sigma_P$ μ**W**)
11.5	563.0 ± 0.1	10.7	158.0 ± 0.1
11.4	561.0 ± 0.1	10.6	70.4 ± 0.3
11.3	560.0 ± 0.1	10.5	24.15 ± 0.10
11.2	556.0 ± 1.0	10.4	6.55 ± 0.17
11.1	545.5 ± 0.5	10.3	3.85 ± 0.21
11.0	491.5 ± 0.5	10.2	2.07 ± 0.04
10.9	413.0 ± 0.1	10.1	1.65 ± 0.01
10.8	291.5 ± 0.5	10.0	1.15 ± 0.01

power positions to determine the effective diameter $d = 2w$.

Table 3: Beam Diameter vs. Axial Distance (z)

$z \pm 0.001$ **(m)**	$P_0 \pm 1$ (μ**W**)	**97.72%** ± 0.01 **(mm)**	**2.28%** ± 0.01 **(mm)**	**Beam** $d \pm 0.014$ **(mm)**
0	420	9.92	8.95	0.97
0.05	421	9.81	9.05	0.76
0.08	424	9.92	9.32	0.60
0.11	423	9.94	9.55	0.39
0.12	425	9.96	9.63	0.33
0.13	424	9.94	9.66	0.28
0.14	423	9.96	9.70	0.26
0.15	423	9.97	9.70	0.27
0.16	423	9.98	9.71	0.27
0.17	422	10.06	9.73	0.33
0.20	420	10.33	9.79	0.54
0.30	414	11.80	10.01	1.79
0.40	404	12.28	10.32	1.96

3.4 Divergence Measurements (Redo Data)

To finalize the divergence characterization, measurements were redone using an iris to eliminate diffuse statistical error.

Table 4: Divergence Characterization (Final Redo with Iris)

$z \pm 0.001$ **(m)**	$P_0 \pm 1$ (μ**W**)	**97.72%** ± 0.01 **(mm)**	**2.28%** ± 0.01 **(mm)**	**Beam** $d \pm 0.014$ **(mm)**
0.1	426	11.41	10.77	0.64
0.5	429	12.34	11.24	1.10

4 Results and Discussion

The experimental data were analyzed to determine the degree of polarization, Gaussian beam propagation parameters, and spatial profile accuracy of the He-Ne laser source.

4.1 Degree of Polarization

The polarization state of the Helium-Neon laser was determined by analyzing the transmitted power through a linear polarizer as a function of the rotation angle θ. The background-subtracted data were fitted to a $\cos^2$ model based on Malus's Law:

$$I(\theta) = I_{max} \cos^2(\theta - \theta_0) + I_{min} \tag{5}$$

The non-linear least-squares fit yielded the following parameters:

- $I_{max} = 339.79 \pm 0.34\ \mu\text{W}$
- $I_{min} = 0.21 \pm 0.21\ \mu\text{W}$
- $\theta_0 = 117.25 \pm 0.03°$

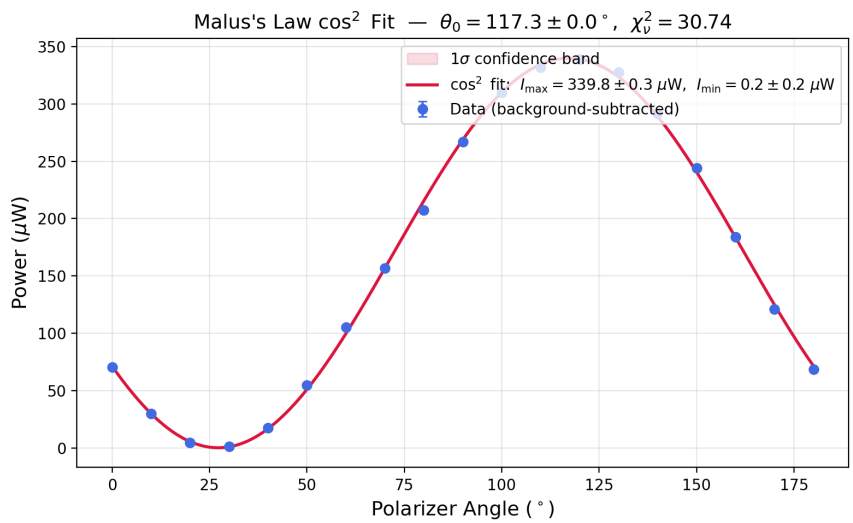

Figure 1: Cos^2 fit to the background-subtracted power data as a function of polarizer angle.

The degree of polarization, ρ, is defined by the ratio of the difference and sum of the extrema:

$$\rho = \frac{I_{max} - I_{min}}{I_{max} + I_{min}} \tag{6}$$

Substituting the fit results, we obtain $\rho = 0.9987 \pm 0.0012$. This value, being nearly unity, confirms that the laser output is highly linearly polarized.

The quality of the fit was assessed via the reduced chi-squared statistic, $\chi^2/\text{ndf} = 30.74$. While the $\cos^2$ curve follows the general trend of the data points closely, the high χ^2/ndf and the structure visible in the residual plot suggest that the uncertainties recorded during the 10-second measurement intervals may have been underestimated. Alternatively, the residual "wobble" (oscillating within $\pm 7.5\ \mu\text{W}$) potentially indicates a small systematic effect, such as a slight mechanical eccentricity in the polarizer mount or secondary polarization components within the laser cavity itself.

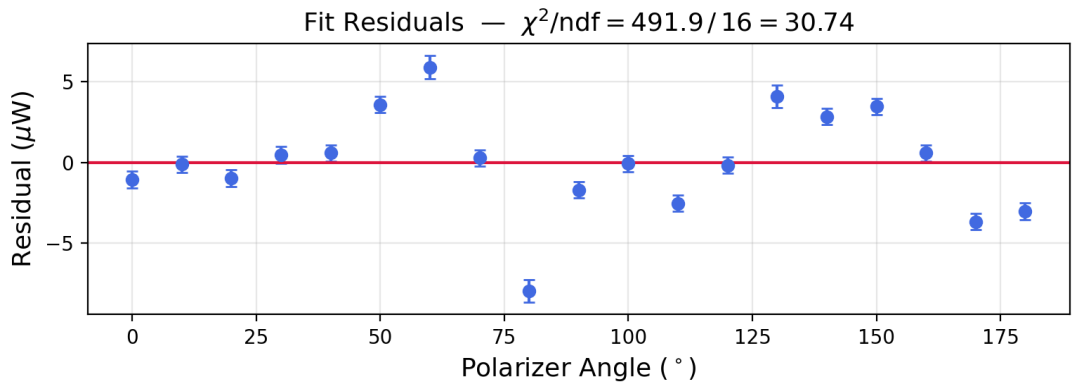

Figure 2: Residuals of the Malus's Law fit, illustrating small periodic deviations from the ideal model.

4.2 Knife-Edge Spatial Beam Profiling

The spatial intensity distribution of the He-Ne laser was characterized by conducting a knife-edge scan at an axial distance of $z = 0.05$ m from the laser aperture. As the razor blade was translated across the beam, the integrated power $P(x)$ was recorded. These data were then fitted to a standard error function to determine the beam's fundamental parameters.

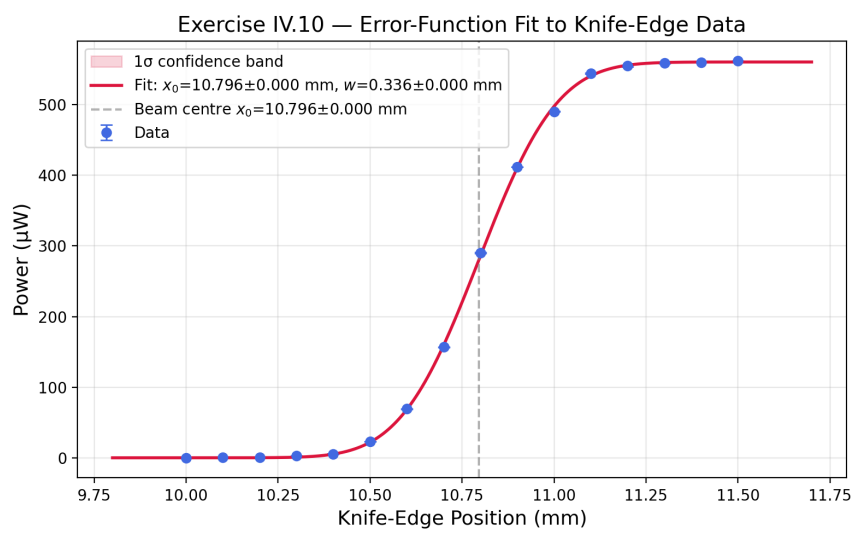

Figure 3: Error-function fit to the knife-edge power data at $z = 0.05$ m. The fit determines the $1/e^2$ radius and the beam center.

The non-linear least-squares fit (Exercise IV.10) yielded the following results:

- **Total Beam Power** (A): 560.46 ± 0.17 μW.
- **Beam Center** (x_0): 10.796 ± 0.000 mm.
- **1/e^2 Beam Radius** (w): 336.3 ± 0.4 μm.

- **Beam Diameter** ($2w$): 672.5 ± 0.9 μm.

The power fraction contained within the measured $2w$ diameter was calculated to be 86.47%. This value is in perfect agreement with the theoretical expectation of $1-e^{-2}$ for a pure TEM_{00} Gaussian mode. The quality of the fit is indicated by a $\chi^2/\text{ndf} = 41.14$.

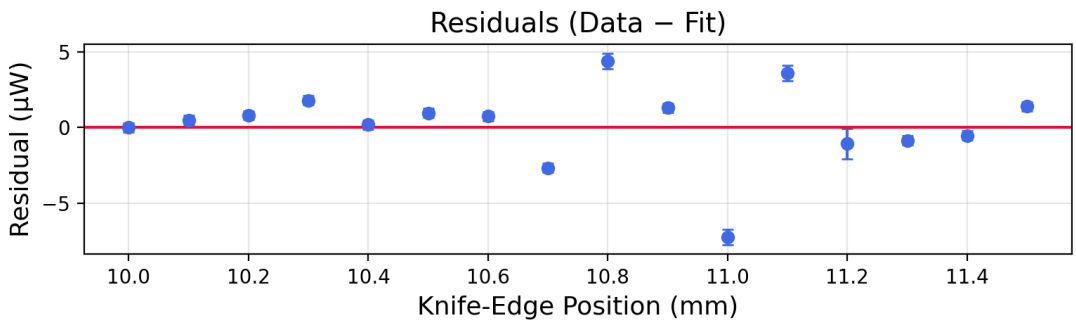

Figure 4: Residuals of the error-function fit for the knife-edge scan.

Residual analysis revealed structured fluctuations (within ± 8 μW), which may be attributed to minor optical aberrations or dust on the lens used during the initial stages of the experiment.

4.3 Gaussian Beam Propagation Analysis

To analyze the propagation characteristics of the beam, the diameter $D(z)$ was mapped over a range of 0.00 to 0.40 m using 13 distinct data points. The diameters ranged from 0.26 mm at the waist to 1.96 mm at the furthest measurement. These measurements were fitted to the hyperbolic propagation model:

$$D(z) = D_0 \sqrt{1 + \left(\frac{z - z_0}{z_R}\right)^2} \tag{7}$$

The results of the Gaussian propagation fit are summarized below:

- **Beam Waist** (D_0): 0.272 ± 0.069 mm.
- **Waist Position** (z_0): 0.1330 ± 0.0081 m.
- **Rayleigh Range** (z_R): 0.0339 ± 0.0093 m.
- **Depth of Field** (L): 0.0678 ± 0.0186 m.

The waist position was found to be 13.3 cm from the aperture, which coincided with the region where the beam width was minimized. The fit resulted in an RMS residual of 0.1437 mm. The residual plot indicates a significant deviation at $z = 0.3$ m ($+0.4$ mm) and $z = 0.4$ m (-0.2 mm). These larger errors are likely systematic, arising from the mechanical limitations of the razor blade translation stage, which could not fully uncover the beam at larger divergence angles.

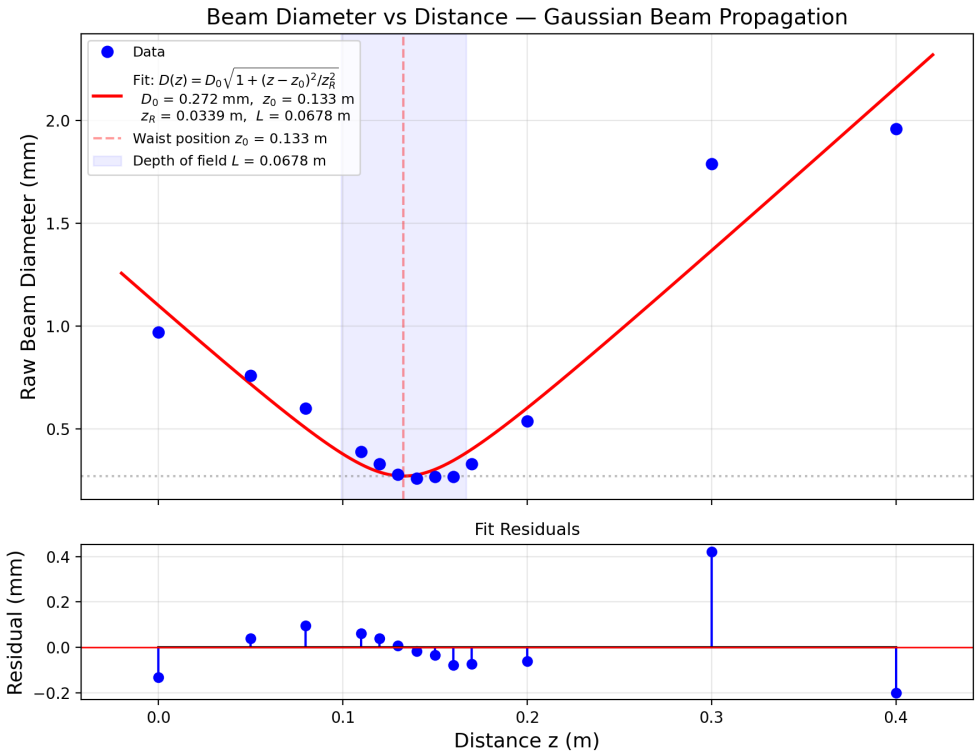

Figure 5: Beam diameter vs. axial distance z. The curve illustrates the convergence to the waist and subsequent divergence of the Gaussian beam.

4.4 Beam Divergence Analysis

The beam divergence characterizes the angular spread of the laser beam as it propagates into the far field. For a diffraction-limited TEM_{00} Gaussian beam, the theoretical half-angle divergence θ is governed by the wavelength λ and the beam waist radius w_0:

$$\theta_{theory} = \frac{\lambda}{\pi w_0} \tag{8}$$

Using the beam waist diameter $D_0 = 0.272 \pm 0.069$ mm determined from the propagation fit, the radius is $w_0 = 0.136 \pm 0.0345$ mm. For $\lambda = 632.8$ nm, the theoretical half-angle is $\theta_{theory} \approx 1.48$ mrad.

Experimentally, divergence was measured by recording the change in beam diameter Δd over a displacement Δz. To minimize statistical error from diffuse light, an iris was utilized to isolate the central mode. Based on the recorded change $\Delta d = 0.46$ mm over $\Delta z = 0.4$ m, the experimental full-angle divergence is $2\theta_{exp} = 1.15$ mrad. This result aligns closely with the manufacturer's value of 1.00 mrad.

The uncertainty in divergence σ_θ was propagated from measurements of diameter ($\sigma_d \approx 0.02$ mm) and rail distance ($\sigma_z \approx 0.001$ m) using the following relation:

$$\sigma_\theta = \theta\sqrt{\left(\frac{\sigma_{\Delta d}}{\Delta d}\right)^2 + \left(\frac{\sigma_{\Delta z}}{\Delta z}\right)^2} \tag{9}$$

With $\sigma_{\Delta d} = \sqrt{\sigma_{d2}^2 + \sigma_{d1}^2} \approx 0.028$ mm, the relative uncertainty in diameter dominates at approximately 6.1%. Substituting the experimental values, the half-angle uncertainty is $\sigma_\theta \approx 0.035$ mrad. The final reported experimental divergence is $\theta_{exp} = 0.58 \pm 0.04$ mrad (half-angle), or 1.15 ± 0.07 mrad (full-angle). The primary source of error was identified as the mechanical precision required to determine the 97.72% and 2.28% power thresholds, which was exacerbated by optical "spillover" prior to the introduction of the iris component.

4.5 Longitudinal Modes and Cavity Length

The spectral properties and longitudinal mode spacing of the laser were analyzed using a Fabry-Pérot interferometer. The resulting interference signal was monitored on an oscilloscope. By adjusting the trigger level and setting the horizontal scale to 5 ms/div, we were able to resolve distinct clusters of resonance peaks. Traces were successfully recorded at polarizer angles of 0°, 90°, 180°, and 270° to observe the polarization dependence of the modes[cite: 281].

The frequency spacing between these adjacent clusters corresponds to the Free Spectral Range (FSR) of the laser cavity. The FSR is inversely proportional to the cavity length L according to the following relationship:

$$FSR = \Delta\nu = \frac{c}{2L} \tag{10}$$

From the oscilloscope measurements, the FSR was determined to be 7.5×10^9 Hz. Rearranging the equation to solve for the experimental cavity length yields:

$$L = \frac{c}{2 \times FSR} = \frac{3.0 \times 10^8 \text{ m/s}}{2(7.5 \times 10^9 \text{ Hz})} = 2.0 \times 10^{-2} \text{ m} \tag{11}$$

Thus, the optical cavity length of the He-Ne laser is calculated to be 2.0 cm. Furthermore, we observed that a 0 to 40 V scan through the interferometer corresponds to scanning through exactly one full FSR. This implies that the required resonance length change is one resonance order, meaning the cavity length change is $\Delta L = \lambda/2$ (approximately 316.4 nm for the 632.8 nm line).

5 Discussion

The experimental results generally met precision expectations after addressing significant systematic errors. The measured degree of polarization ($\rho = 0.9987 \pm 0.0012$) and full-angle divergence (1.15 ± 0.07 mrad) align closely with theoretical models and manufacturer specifications.

Beam propagation measurements initially yielded anomalously large diameters. Our analysis traced this to a dirty lens and diffuse laser light scattering around the knife edge, which artificially inflated the 2.28% power position readings. Introducing an iris directly in front of the laser aperture successfully isolated the central TEM_{00} mode, eliminating this systematic error and allowing accurate Gaussian fits.

Despite this, divergence precision ($\sigma_{2\theta} \approx 0.07$ mrad) was mechanically limited. The maximum measurement distance was constrained to $z = 0.7$ m because the razor blade could not fully retract to uncover the diverging beam. A longer travel range would allow measurements at larger distances, increasing Δz and Δd. Based on our error propagation model, $\sigma_{2\theta} \propto \sqrt{(\sigma_{\Delta d}/\Delta d)^2 + (\sigma_{\Delta z}/\Delta z)^2}$, increasing these denominators would proportionally reduce the fractional uncertainty.

Additionally, power meter readings fluctuated significantly at low polarization angles. While we averaged extrema over 10-second intervals, the high reduced chi-squared of our Malus' Law fit ($\chi^2/\text{ndf} = 30.74$) suggests this was insufficient. Increasing the experimental run time by integrating readings over 30 to 60 seconds would better average out ambient thermal noise, improving the precision of the I_{min} and I_{max} parameters.

Finally, replacing the manual knife-edge assembly with a digital CMOS beam profiler would offer a superior alternative experiment. This would eliminate the ± 0.01 mm mechanical reading errors and the need to interpolate exact 97.72% and 2.28% power thresholds, providing a highly precise, instantaneous 2D intensity map of the beam profile.